Quantification: 543_z13
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	11
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1862	1736	823.012	px^2
major_axis_length_px	lacuna shape	78.3615	74.0218	22.3719	px
minor_axis_length_px	lacuna shape	30.8137	31.6227	6.9101	px
aspect_ratio	lacuna shape	2.5873	2.6086	0.6752	unitless
eccentricity	lacuna shape	0.9006	0.9236	0.0676	unitless
solidity	lacuna shape	0.9037	0.9231	0.0388	unitless
roots_count	headline	4.3636	4	2.1574	count
ring_length_r30_px	headline	310.364	331	59.7533	px
ring_length_r60_px	reported	922.818	912	141.285	px
owned_length_px	ownership-dependent	2884.72	2192.32	1747.82	px
edge_count	ownership-dependent	126.727	107	83.8357	count
mean_edge_length_px	ownership-dependent	24.3069	22.7652	3.6281	px
perimeter_px	normalised	207.479	205.037	54.0901	px
ring_area_r30_px2	normalised	8447.82	8219	1466.9	px^2
ring_area_r60_px2	normalised	21999.7	21794	3275.21	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0374394	0.035343	0.00859746	px^-1
ring_density_r60	normalised	0.042466	0.0434941	0.00710373	px^-1
roots_per_100px_perimeter	normalised	2.1027	2.1226	0.8621	per 100 px
ring_attached_length_r30_px	network v2	244.909	260	60.881	px
ring_attached_length_r60_px	network v2	657.909	740	203.508	px
ring_length_w_r30_px	network v2	352.242	376.534	66.4289	px
ring_length_w_r60_px	network v2	1046.49	1055.01	158.498	px
sholl_crossings_r10	network v2	6.8182	7	2.2724	count
sholl_crossings_r20	network v2	11	12	2.7203	count
sholl_crossings_r30	network v2	13.4545	13	2.1616	count
ring_width_mean_r30_px	width	7.0667	7.3794	0.6515	px

Per field
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measure	value	units
analysed_area_px2	1.02809e+06	px^2
total_skeleton_length_px	41364	px
canalicular_length_density_per_px	0.0402337	px^-1
junction_count	926	count
junction_density_per_px2	0.0009007	px^-2
lacuna_count	11	count
lacunae_per_px2	1.07e-05	px^-2
skeleton_component_count	231	count

mean_component_length_px	179.065	px
median_component_length_px	24	px
field_length_density_w_per_px	0.0457041	px^-1
field_density_without_flagged_per_px	0.0402337	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	256	px
unattached_component_count	13	count
canalicular_length_density_attached_per_px	0.0402337	px^-1
field_length_density_with_unattached_per_px	0.0404827	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6176	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10	px
width_pixels_used	41364	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2019	79.1535	33.2822	2.3783	0.9073	0.9417	0.4378	97.9678	270.286	False
2	839	61.6665	17.863	3.4522	0.9571	0.906	0.068	131.349	828.439	False
3	1736	73.3356	33.3815	2.1969	0.8904	0.8374	0.2312	218.82	919.248	False
4	1138	46.8317	31.6227	1.481	0.7376	0.9252	0.396	263.104	340.498	False
5	2267	87.9083	33.2852	2.6411	0.9255	0.9473	0.0042	459.621	481.093	False
6	860	45.4745	26.4762	1.7176	0.813	0.854	0.07	475.221	748.086	False
7	2150	103.918	27.5311	3.7745	0.9643	0.9231	0.2711	546.673	873.734	False
8	3069	109.359	37.084	2.9489	0.9407	0.9099	0.049	642.67	188.33	False
9	1616	74.0218	28.3763	2.6086	0.9236	0.9245	-0.0477	731.536	630.238	False
10	1438	72.5092	25.5367	2.8394	0.9359	0.9236	0.0953	793.137	478.379	False
11	3350	107.799	44.5116	2.4218	0.9108	0.8475	-0.088	854.917	288.104	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	5	295	912	3742.82	172	21.7606	7.4007
2	4	354	951	4644.11	204	22.7652	6.6482
3	8	331	876	1524.6	62	24.5903	6.4844
4	2	238	858	1293.3	52	24.8712	7.7768
5	6	347	1047	5504.27	247	22.2845	7.8264
6	3	353	942	2173.93	107	20.3171	7.3794
7	4	337	1052	2914	130	22.4154	6.3621
8	3	274	875	2192.32	87	25.1991	7.4497
9	3	230	748	1612.2	53	30.4188	6.815
10	2	239	694	535.144	17	31.4791	5.9049
11	8	416	1196	5595.24	263	21.2747	7.6863

Normalised per lacuna

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lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	205.037	8499	22530	1	1	0.03471	0.0404794	2.4386
2	151.297	6982	17707	1	1	0.0507018	0.0537076	2.6438
3	205.037	8219	20045	1	1	0.0402725	0.0437017	3.9017
4	139.681	6734	18999	1	1	0.035343	0.0451603	1.4318
5	223.823	8900	23453	1	1	0.0389888	0.0446425	2.6807
6	141.338	6516	17817	1	1	0.0541743	0.0528708	2.1226
7	254.936	9810	24446	1	1	0.0343527	0.0430336	1.569
8	278.593	10387	26403	1	1	0.0263791	0.0331402	1.0768
9	195.723	8083	21794	1	1	0.0284548	0.0343214	1.5328
10	184.995	7835	21305	1	1	0.0305042	0.0325745	1.0811
11	301.806	10961	27498	1	1	0.0379527	0.0434941	2.6507

Network v2 per lacuna

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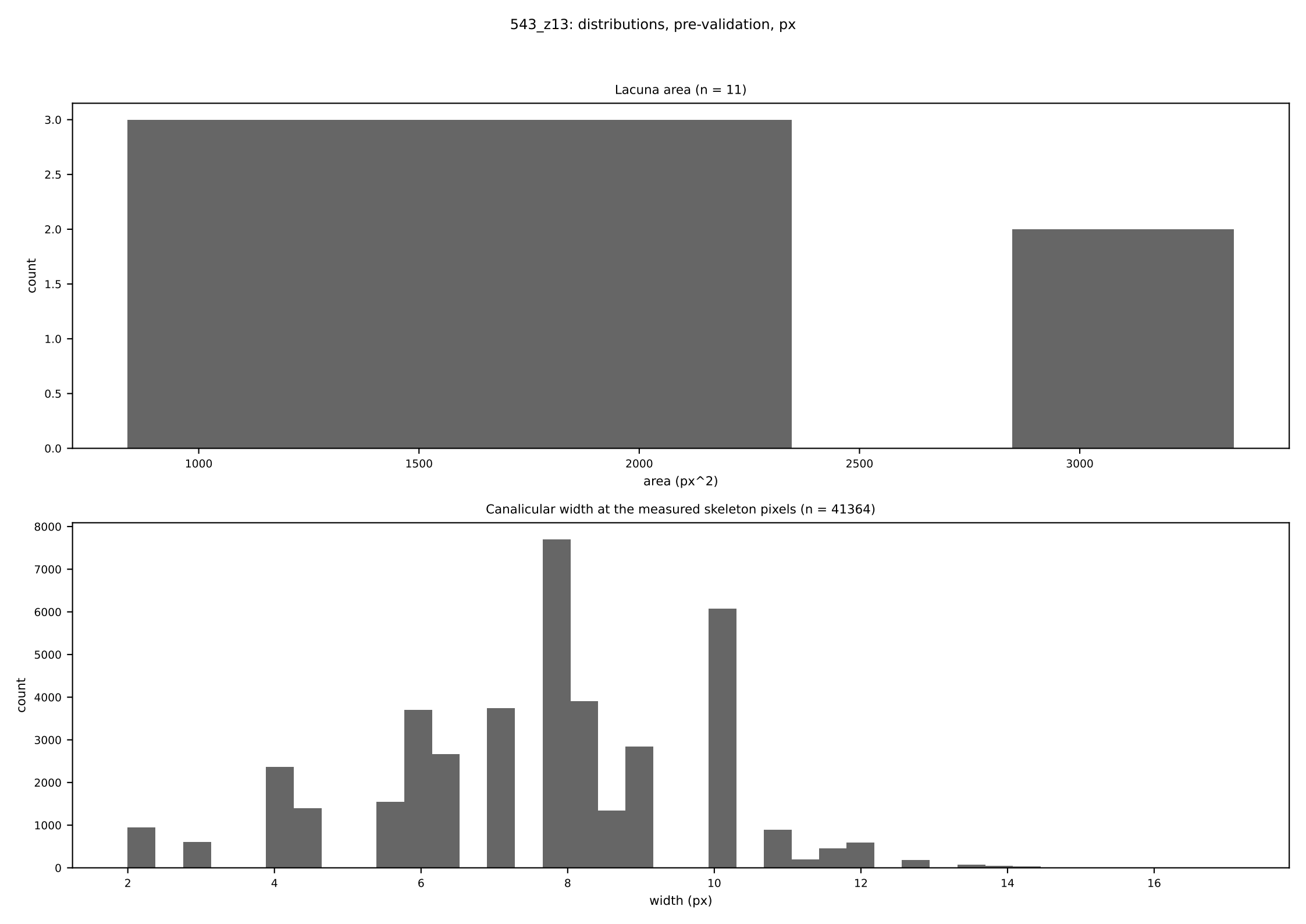
lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	274	826	353.889	1057.05	9	11	10
2	274	740	391.279	1062.28	6	13	14
3	260	523	376.534	993.749	11	13	12
4	147	460	276.108	972.151	4	8	13
5	286	784	389.321	1168.36	7	12	17
6	310	760	392.522	1055.01	5	13	15
7	248	503	380.463	1188.97	7	12	13
8	228	855	315.907	1017.42	8	10	15
9	154	496	260.652	856.838	4	8	12
10	183	324	263.752	778.713	5	6	11
11	330	966	474.233	1360.82	9	15	16


543_z13: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 41364)
count
width (px)